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Are Head-mounted Displays Really Not Suitable for Driving Simulation? A Comparison with a Screen-Based Simulator

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Abstract—Head-mounted displays (HMDs) are considered a promising, highly immersive display technology, which has been widely discussed in the context of driving simulation. The literature suggests negative effects of HMD use on simulator sickness and positive effects on perception and presence. However, the reported simulation scenarios with outdated HMD technologies are very different according to today's standard, which makes it difficult to draw a final conclusion. Therefore, in the present study a comparison between a modern HMD with a screen-based (LED wall) simulator is made in a repeated-measures study with $n = 31$ subjects. In contrast to previous work, there was no clear evidence for differences between the two simulators in distance and speed perception, simulator sickness, or presence. Most of the evidence in fact suggests equivalence across the two simulators. Accordingly, the present study does not identify any disadvantages of using current HMDs in driving simulation.

Index Terms—driving simulation, virtual reality, head-mounted display, simulator sickness, presence, perception

I. INTRODUCTION

In driving simulation the goal is usually to provide the driver with a realistic impression of the displayed scene. Human perception and information processing are central in this context. A driving simulator is required to correctly reproduce sensory inputs received by the driver [1]. [2] goes so far as to say that evaluating driving behavior is only sensible if the sensations received in the driving simulator correspond with those which would be received in reality.

Visual information is key to the driving task [3]; **distance** [4] and **speed perception** [5] are especially important as they are required for tasks such as stopping, overtaking, or maintaining speed. Both are based on visual input information [6]. Regarding distance perception, pictorial cues and binocular cues can be distinguished. While pictorial distance cues, such as the relative size of objects or occlusion, are comparatively

easy to implement, binocular cues are more difficult to reproduce. Convergence is a binocular depth cue which refers to the angle at which the eyes are relative to each other when focusing [7]. However, this is thought to be relevant only at very short distances of 2-3 m. Accommodation refers to the contraction of the ciliary muscle. In a moving environment, also motion-dependent cues influence depth perception, such as motion parallax: Objects which are closer to the observer seem to move faster than objects in a far distance [7].

When it comes to speed perception, optic flow is believed to be the main determining variable. It is generated by the motion of an observer and his surroundings. When driving and looking in the direction of travel, environment objects seem to move out of the central field of view (FOV) into periphery [7]. The faster objects move along the optic flow, the faster the intrinsic movement is perceived. Summarizing the information presented above, visual perception undoubtedly determines driving behavior. Driving simulators should replicate real-world driving behavior as good as possible (also referred to as *behavioral validity* [8]), as only then it is possible to draw inferences on the real-world context.

Visual perception is largely determined by *simulator fidelity*, commonly understood as the capability of the simulator to reproduce the real-world driving context [8]. Display hardware can contribute to the correct representation of real-world cues or its deterioration [6]. Both software and hardware elements determine visual fidelity. As the present study is aimed at investigating influences of the display type, we will focus on effects of hardware technology.

A. Display technologies in driving simulators

Among the most frequently used display solutions in driving simulators are screens (mostly LCD panels), projectors, or

head-mounted displays (HMDs). HMDs offer a cheap solution as all elements of a real vehicle can be displayed virtually, hence no physical mockup is required. [9] proposed that 40 % of the studies investigated could be conducted using HMDs. They mention as a benefit that HMDs are, at least partially, able to provide binocular depth cues. Perception may thus be enhanced with HMDs. McIntire et al. concluded in their 2014 review [10] that 60 % of the studies show beneficial effects of binocular displays. However, [11] proclaim that the mere fact that a three-dimensional world is transferred to a two-dimensional sort of display always leaves room to ambiguity in sensory cues. Even state-of-the-art stereoscopic displays are incapable of delivering an appropriately variable focus, and thus, provide vergence cues without an adequate accommodation, causing a vergence-accommodation conflict.

[9] also state that the **sense of presence** should be increased with HMDs, which was also reported by [12]. The sense of presence is defined as the sense of being there in a virtual environment [13]. It is assumed to depend on immersion, which in turn is understood as the degree to which the real-world is shut out by a setup [14]. The higher the sense of presence, the more realistic drivers are expected to behave within the virtual environment [12]. The literature on effects of using HMDs on presence is ambiguous, however, as effects could not be confirmed by all studies [15], [16].

The incidence of **simulator sickness** is often reported to be increased with HMDs [12], [16]–[18], forming a potential disadvantage. Simulator sickness is a physical discomfort that can occur when the cues received in a virtual environment do not match the expectations of the receiver [19]. Sickness is to be reduced to minimize dropouts and to preserve validity [20], [21]. Also regarding effects on simulator sickness, effects of HMDs have not been fully elucidated. [15] and [22] could not confirm any effects of using HMDs on simulator sickness. [23] even found increased simulator sickness in projection screens when compared to HMDs.

Besides displays effects, there are some practical implications when using HMDs. First, when wearing HMDs, participants are always exposed to extra weight and often extra cables, which can lead to an unnatural feeling and less head movements. Second, the FOV is theoretically always limited with HMDs (approx. 110° with most to-date HMD solutions, approx. 200° in the real world [24]), potentially reducing optic flow, while peripheral information is available with other displays. We also found that many research papers on HMDs are comparatively old or used low-fidelity HMDs, which is often associated with lower image resolution [9], contrast, FOV, etc. Thus, the differences found between HMDs and other displays may also be attributed to different hardware properties. The horizontal FOV, for instance, was reported to influence presence [25], simulator sickness [23], [26], [27], distance and speed perception [6], [28], [29]. Image resolution can influence distance perception by atmospheric depth [30]. Optic flow should depend on the horizontal FOV [7].

Another limitation of previous studies is that there were differences in the display technology compared (projection:

[15], [16], [23]; displays: [12], [31]). When multiple screens are combined, display boundaries are usually visible, which reduces the immersion of the visualization. Projection systems, on the other hand, are often low contrast, which can also have a negative impact on immersion. In the present study, we therefore compare the HMD to a highly immersive and novel visualization technology in driving simulation: the LED wall.

LED walls have been previously used especially in advertisement but are today also used in television studios. Smearing effects can be reduced at low latency and realistic blending can be realized, which is a common issue with projectors. Furthermore, due to the high image contrast of the LED technology, it is possible to reproduce colors and lighting conditions realistically.

B. Research gap

In summary, HMDs offer a cheap and easy-to-implement solution for visualizing driving scenes and thus hold great potentials for the area of driving simulations. Findings on effects of using HMDs, however, are so far ambiguous and require clarification. Positive and negative effects of HMDs have been proclaimed but were often not replicated. However, studies that did not find effects of HMDs have not provided statistical evidence for the equivalence of different visualization systems either. Using Bayesian hypothesis testing instead of frequentist statistics (see section "Statistical analysis"), we aim to also address statistical equivalence in the present study to identify whether there really is no disadvantage in using HMDs.

Many studies have been conducted with outdated HMDs, which were also disadvantageous in terms of hardware aspects beyond the display type. For the reasons outlined above, we conducted a new comparative study of a state-of-the-art HMD with an LED wall. As described above, the LED wall technology is rather new in the context of driving simulation and has therefore not been included in previous investigations on effects of the display type.

C. The present study

A within-subject study was conducted in which subjects experienced an LED wall simulator and an HMD simulator in succession in different driving scenarios. We applied the following hypotheses to the study:

- H1:** Distance perception is more accurate in the HMD simulator due to the availability of binocular depth cues.
- H2:** Speed perception is more accurate in the LED wall simulator due to a larger FOV and peripheral vision.
- H3:** The sense of presence is higher in the HMD simulator.
- H4:** Incidence and severity of simulator sickness are higher in the HMD simulator.

II. METHODOLOGY

A. Sample

Out of the initial sample, 22 participants dropped out, 6 of them for technical reasons. Another 16 participants dropped out due to simulator sickness; 9 while using the HMD

simulator, and another 7 while using the LED wall simulator. 31 participants completed the study, of which 3 were female and 28 were male. Participants were aged between 20 and 57 years, with an average age of 31 years.

B. Materials

The software Unreal [32] was used for visualization. Object and agent information and trajectories were provided by proprietary software modules to the graphics engine. Hardware components as well as software integration of the steering wheel and pedals were largely standardized among the mock-ups. The frame rate was limited to constant 60 Hz in both simulators to avoid fluctuation and the achieved contrast of the HMD using OLED technology is comparable to that of the LED wall (approx. 1,000,000:1). A limitation was that the brake in the VR simulator worked using a force sensor while the brake in the LED wall simulator worked using a stroke sensor. The brakes were manually calibrated to ensure the same braking response and effect in both simulators.

The **HMD driving simulator** (Figure 1) consisted of a seat box built from a BMW 5-series vehicle mockup. A matching G30 design model was displayed in the virtual scene. For visualization, the Varjo VR-3 was used [33]. According to Varjo's own statement [33], the VR-3 currently offers the highest resolution as well as one of the widest FOVs (1920×1920 px per eye and 115°) on the market. The VR-3 should therefore deliver the fairest possible comparison. Including the head-band, the VR-3 has a total weight of 944 g. The visual representation of limbs was not realized in the given setup, therefore participants were unable to see their own hands. The **LED wall simulator** (Figure 1) was equipped with a full-vehicle BMW 5-series mockup. The simulator was equipped with a small motion system, which was disabled to allow comparability with the static HMD setup. An LED wall with a horizontal FOV of 310 was used, with an overall resolution of approximately 26 pixels per degree. Rear vision was implemented with LCD display mirrors.



Fig. 1. Illustration of the two simulators. HMD = left, LED wall = right.

C. Design

The present study was performed as a 2×4 within-subjects design including the factors driving simulator (HMD vs. LED

wall) and scenario (country road, highway, test track, urban). The country road scenario was always presented first. The remaining scenarios (highway, test site, urban) were presented in a counter-balanced order. The order of presentation was the same in the two driving simulators (for each participant). The order of the driving simulators was also balanced to prevent order effects.

1) *Dependent variables:* Distance and speed perception were measured by verbally provided open estimates. Simulator sickness was measured using the single-item Misery Scale (MISC) [34], [35]. The MISC is a progressive scale which was presented combined with a color-indicator to enhance comprehensibility [21]. Presence was measured using an adapted version of the Slater-Usch-Steed presence questionnaire ([36], see [37] for the adapted version).

2) *Statistical analysis:* All hypotheses were formulated in a directed manner on the basis of the existing literature; however, as already explained, the study situation was heterogeneous and partially contradictory, respectively. To the best of our knowledge, the present study is also the first one to compare an HMD with an LED wall simulator. It thus has an exploratory character to a certain extent, since it would also be plausible that there is no difference between the two display types. To account for this, all results will be evaluated based on Bayesian hypothesis testing. In contrast to the frequentist approach, Bayesian statistics allow statistical support for both difference and equivalence hypotheses. The Bayes factor can be considered as a relative indicator of the probability of the observed data given equivalence or an effect. Jeffreys provided recommendations on the interpretation of Bayes factors [38]. A Bayes factor between 1 and 3 is considered weak evidence, a Bayes factor between 3 and 10 is moderate evidence, and a Bayes factor > 10 is considered strong evidence for the effect model. A Bayes factor of 1/3 can be considered weak evidence, a Bayes factor of 1/10 moderate evidence, and a Bayes factor of 1/30 strong evidence for the null model [39].

D. Procedure

The drives were performed subsequently with short questionnaires after each scenario. The **country road** scenario served as an introductory drive. The **highway** scenario included different tasks related to distance and speed perception. The distance-related tasks followed a three-stage scheme, in which there was always a vehicle driving in front. In the first task (*blind following task*), the vehicle in front was driving at a speed of 92 km/h and participants were instructed to follow it in a constant distance of 50 m and to let the experimenter know when they thought to have the correct distance. The actual distance was considered as an indicator of *active*, i.e. perception with regulation effort, *distance perception*. Second (*blind distance keeping task*), participants were supported to achieve the correct distance of 50 m, which was achieved by verbal instructions of the experimenter, and were then instructed to keep this distance as accurately as possible for one minute. No speedometer information was available in the first two tasks. The standard deviation of the distance

to the vehicle in front was considered as an indicator of *active distance perception*. In the third task (*acceleration task*), speedometer information was available. The third task was instructed exactly the same way as the second, with the only difference that the driving speed of the vehicle in front changed from an initial speed of 92 km/h to 105 km/h. The point of ego-vehicle acceleration over 95 km/h was considered as an indicator of *active speed perception*. For the *passive speed perception task*, again no speed information was available to the participants. An automated drive was carried out with a driving speed of 137 km/h and participants were asked to estimate the driving speed. The provided estimate was considered as a measure of *passive speed perception*. In the **test site** scenario, again distance perception was investigated. First (*egocentric distance perception task*), participants were asked to estimate their egocentric distance to a person placed next to a cone in a distance of 50 m, 70 m, or 120 m, with each distance presented twice in a randomized order. A cone at a distance of 20 m was always given as a reference. The distance estimates are indicators of *passive distance perception*. Second, there was a *target braking task*, in which participants were asked to come to stop at a designated position marked by cones and an adjacent vehicle. The distance to the stopping line was interpreted as an indicator of *active distance perception*. Last, in the *urban task*, participants were navigated through a large city on specific routes. The urban scenario included different types of road users, such as other vehicles or bicyclists, and active traffic signals. Typical of urban traffic, it included a lot of stopping and turning. Sickness and presence were assessed after each scenario.

III. RESULTS

All Bayesian analyses were performed using the software JASP [40]. Default priors [41] were used for the analysis, as based on the mixed study situation no concrete priors could be formulated. To assess the robustness of the results, all analyses were repeated with different informed priors [42]. An annotated results file including the robustness checks as well as more detailed information on the performed analyses will be published on the Open Science Framework (OSF; <https://osf.io>) upon acceptance of this paper, as recommended by [39].

A. Perception

In the blind following task, the Bayes factor analysis indicated moderate evidence for equivalence across the two driving simulators regarding the distance to the vehicle in front $BF_{incl} = 0.26$, Figure 2. In the blind distance keeping task, moderate evidence was found for equivalence across the two simulators in the standard deviation of the distance to the vehicle in front, $BF_{incl} = 0.27$, Figure 3. The analysis of the acceleration task indicated moderate evidence for equivalence in the time to react to the accelerating vehicle in front, $BF_{incl} = 0.26$, Figure 4. In the passive speed perception task, the analysis yielded weak evidence for equivalence across the simulators, $BF_{incl} = 0.74$, Figure 5. The analysis on the accu-

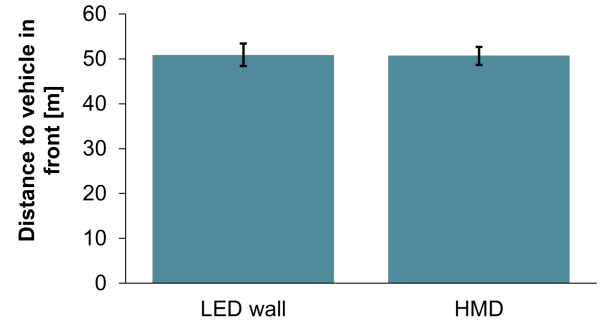


Fig. 2. Descriptive statistics for the blind following task. Error bars indicate standard errors of the mean.

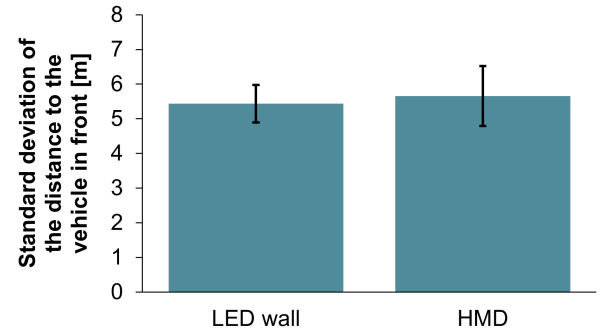


Fig. 3. Descriptive statistics for the blind distance keeping task. Error bars indicate standard errors of the mean.

racy of the passive distance estimates provided throughout the egocentric distance perception task was performed including the factor trial in the null model, as there was no hypotheses on the impact of trial. The analysis indicated weak evidence for equivalence across the simulators for the 50 m distance, $BF_{incl} = 0.61$, the 70 m distance, $BF_{incl} = 0.37$, and the 120 m distance, $BF_{incl} = 0.32$ Figure 6. In the target braking task, there were three trials, yet the effect of trial was again not relevant to the specified hypotheses and was therefore included in the null model. The analysis indicated weak evidence for a worse performance in the HMD simulator, $BF_{incl} = 1.87$, Figure 7.

B. Simulator sickness

The simulator sickness analysis indicated moderate evidence for the absence of an effect of the driving simulator, $BF_{incl} = 0.24$, and strong evidence for an effect of the driving scenario, $BF_{incl} = 1156.81$ (Figure 8). Bayesian post-hoc tests pointed out that significantly more simulator sickness was induced in the urban scenario compared to all other scenarios. There was no interaction of visualization technology and scenario, $BF_{incl} = 0.09$ (Figure 8).

C. Presence

Presence was analyzed including the factor driving scenario. The analysis indicated moderate evidence for the absence of an effect of the factor driving simulator, $BF_{incl} = 0.24$, strong

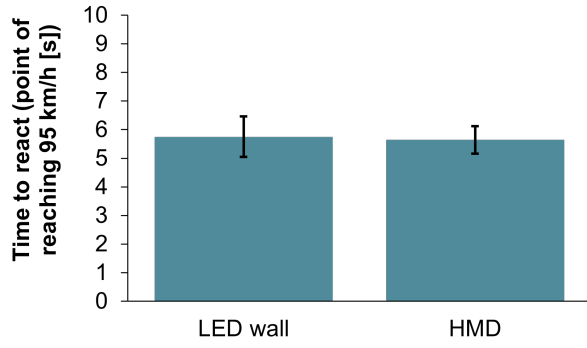


Fig. 4. Descriptive statistics for the acceleration task. Error bars indicate standard errors of the mean.

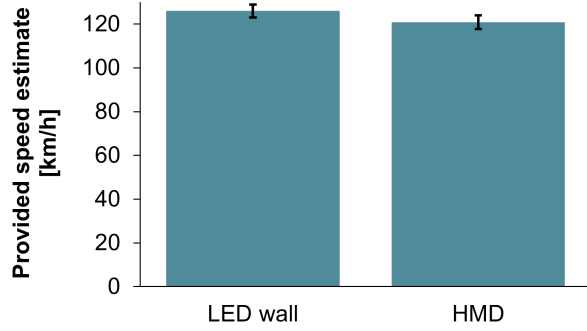


Fig. 5. Descriptive statistics for the passive speed perception task (highway). Error bars indicate standard errors of the mean.

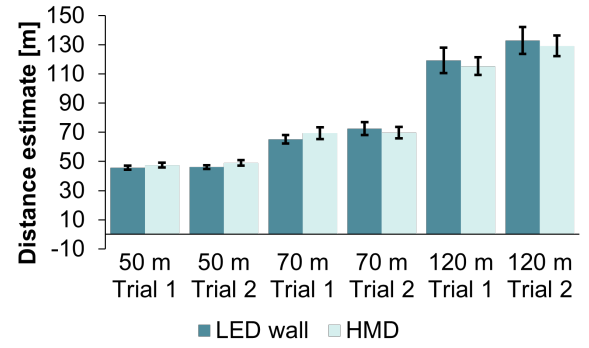


Fig. 6. Descriptive statistics for the egocentric distance perception task. Error bars indicate standard errors of the mean.

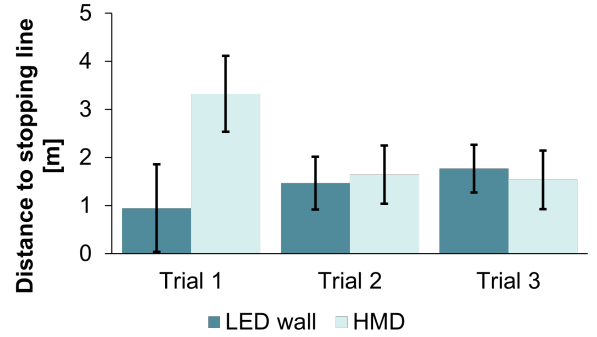


Fig. 7. Descriptive statistics for the target braking task. Error bars indicate standard errors of the mean.

evidence for the absence of an effect of the driving scenario, $BF_{incl} = 0.07$, and strong evidence for the absence of an interaction of simulator and scenario, $BF_{incl} = 0.04$ (Figure 9).

IV. DISCUSSION

In the present study, an HMD and an LED wall simulator were compared with regard to possible differences in perception, simulator sickness, and presence. Contrary to our expectations, we did not observe any positive effects of HMD use on distance perception and no negative effects on speed perception. There was moderate evidence for equivalence across the two simulators in the blind following and distance keeping tasks and in the acceleration task. In the passive speed perception and the egocentric distance perception tasks, we found anecdotal evidence for equivalence. In the target braking task, there was anecdotal evidence for a better perception in the LED wall simulator. The distance estimation tasks on the highway and the egocentric distance perception tasks were mainly concerned with longer distances, where binocular cues tend not to play a role. For this reason, the stereoscopic presentation in the HMD hardly provides an advantage here, which is why the absence of an effect is not too surprising, even though positive effects might have been expected due to the lower distance to the display medium in the HMD simulator. The better performance in the LED wall simulator in the braking task

is rather surprising considering that small distances (< 10 m) matter here (therefore also binocular cues matter) and the display is closer in the HMD simulator, reducing the conflict of focusing distance (display) and presented distance. However, the evidence was only anecdotal.

Regarding speed perception, positive effects of the LED wall simulator were expected due to the wider FOV, which could not be confirmed. [6] suggested that a 120° horizontal FOV would be required to correctly perceive speeds. The HMD simulator offered a FOV (115°) that was already very close to this recommendation, so the difference may simply not have been large enough to make an impact.

There was evidence for equivalence across the two simulators with regard to the incidence and severity of simulator sickness, which did not match our initial expectations. However, other authors could neither identify any negative effects of HMD usage [15], [22], [23]. Furthermore, the present study, as outlined throughout the introduction, used a state-of-the-art high-fidelity HMD, which was not the case in most previous investigations. Effects found in earlier studies could also be attributable to other properties of the used HMDs, such as low image resolution. There was also a scenario effect: simulator sickness in the urban scenario was significantly higher compared to the other scenarios, replicating [21].

When considering the sense of presence, we found proof of

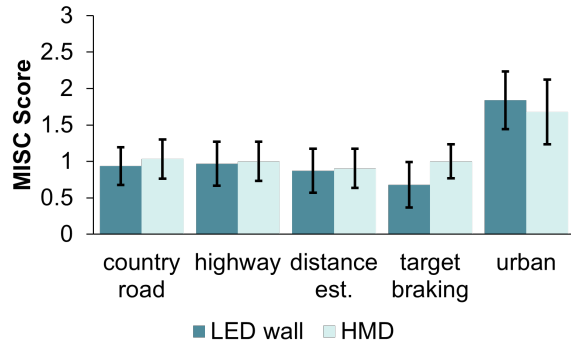


Fig. 8. Descriptive statistics for simulator sickness. Error bars indicate standard errors of the mean.

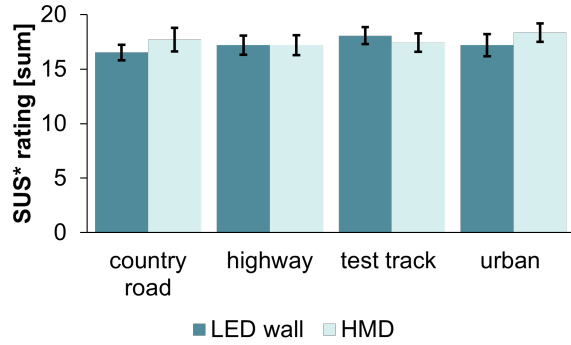


Fig. 9. Descriptive statistics for the sense of presence. Error bars indicate standard errors of the mean.

equivalence across the two driving simulators, contradicting our expectations of a higher sense of presence in the HMD simulator. There were several other studies that likewise did not find any positive effects of HMD use on presence [15], [16]. Furthermore, the present study is the first to compare an HMD against an LED wall. The LED wall technology is considered highly immersive, especially when compared to LCD panels and projection systems. Accordingly, the equivalence across the two simulators could be a result of a particularly high experience of presence in the LED wall simulator. The full vehicle mockup used in the LED wall simulator may also influence presence [43]. This highly familiar vehicle environment might have contributed to participants experiencing a high sense of presence in the LED wall simulator, especially since other studies in which a positive effect of the HMD on presence was found did not use a full vehicle mockup in their comparison condition [12].

V. CONCLUSION

Summarizing, the present study could not identify any clear evidence for differences, and in most cases evidence of equivalence instead, across the HMD and the LED wall simulator, considering the factors perception, sickness, and presence. The results are rather surprising in light of the listed literature. Possible reasons have been outlined throughout the

discussion. A remaining problem of previous and the present study is that it is hardly possible to causally attribute effects (or no effects) to the display type itself. Display hardware always differs in several aspects (e.g., resolution, image contrast, ...), some of which could have unknown, possibly interacting and counteracting effects. Although the causal interpretation should therefore be treated with caution, we conclude that the present study does not provide any evidence of negative implications of using HMDs in driving simulation.

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